

Exercises and Problems for Section 14.3

Exercises

In Exercises 1–8, find the equation of the tangent plane at the given point.

1. $z = ye^{x/y}$ at the point $(1, 1, e)$
2. $z = \sin(xy)$ at $x = 2, y = 3\pi/4$
3. $z = \ln(x^2 + 1) + y^2$ at the point $(0, 3, 9)$
4. $z = e^y + x + x^2 + 6$ at the point $(1, 0, 9)$
5. $z = \frac{1}{2}(x^2 + 4y^2)$ at the point $(2, 1, 4)$
6. $x^2 + y^2 - z = 1$ at the point $(1, 3, 9)$
7. $x^2y^2 + z - 40 = 0$ at $x = 2, y = 3$
8. $x^2y + \ln(xy) + z = 6$ at the point $(4, 0.25, 2)$

In Exercises 9–12, find the differential of the function.

9. $f(x, y) = \sin(xy)$
10. $g(u, v) = u^2 + uv$
11. $z = e^{-x} \cos y$
12. $h(x, t) = e^{-3t} \sin(x + 5t)$

In Exercises 13–16, find the differential of the function at the given point.

13. $g(x, t) = x^2 \sin(2t)$ at $(2, \pi/4)$
14. $f(x, y) = xe^{-y}$ at $(1, 0)$
15. $P(L, K) = 1.01L^{0.25}K^{0.75}$ at $(100, 1)$
16. $F(m, r) = Gm/r^2$ at $(100, 10)$

Problems

17. At a distance of x feet from the beach, the price in dollars of a plot of land of area a square feet is $f(a, x)$.
 - (a) What are the units of $f_a(a, x)$?
 - (b) What does $f_a(1000, 300) = 3$ mean in practical terms?
 - (c) What are the units of $f_x(a, x)$?
 - (d) What does $f_x(1000, 300) = -2$ mean in practical terms?
 - (e) Which is cheaper: 1005 square feet that are 305 feet from the beach or 998 square feet that are 295 feet from the beach? Justify your answer.
18. A student was asked to find the equation of the tangent plane to the surface $z = x^3 - y^2$ at the point $(x, y) = (2, 3)$. The student's answer was

$$z = 3x^2(x - 2) - 2y(y - 3) - 1.$$
 - (a) At a glance, how do you know this is wrong?
 - (b) What mistake did the student make?
 - (c) Answer the question correctly.
19. (a) Check the local linearity of $f(x, y) = e^{-x} \sin y$ near $x = 1, y = 2$ by making a table of values of f for $x = 0.9, 1.0, 1.1$ and $y = 1.9, 2.0, 2.1$. Express values of f with 4 digits after the decimal point. Then make a table of values for $x = 0.99, 1.00, 1.01$ and $y = 1.99, 2.00, 2.01$, again showing 4 digits after the decimal point. Do both tables look nearly linear? Does the second table look more linear than the first?
 - (b) Give the local linearization of $f(x, y) = e^{-x} \sin y$ at $(1, 2)$, first using your tables, and second using the fact that $f_x(x, y) = -e^{-x} \sin y$ and $f_y(x, y) = e^{-x} \cos y$.
20. Find the local linearization of the function $f(x, y) = x^2y$ at the point $(3, 1)$.
21. For the differentiable function $h(x, y)$, we are told that $h(600, 100) = 300$ and $h_x(600, 100) = 12$ and $h_y(600, 100) = -8$. Estimate $h(605, 98)$.
22. (a) Find the equation of the plane tangent to the graph of $f(x, y) = x^2e^{xy}$ at $(1, 0)$.
 - (b) Find the linear approximation of $f(x, y)$ for (x, y) near $(1, 0)$.
 - (c) Find the differential of f at the point $(1, 0)$.
23. Find the differential of $f(x, y) = \sqrt{x^2 + y^3}$ at the point $(1, 2)$. Use it to estimate $f(1.04, 1.98)$.
24. (a) Find the differential of $g(u, v) = u^2 + uv$.
 - (b) Use your answer to part (a) to estimate the change in g as you move from $(1, 2)$ to $(1.2, 2.1)$.
25. An unevenly heated plate has temperature $T(x, y)$ in $^\circ\text{C}$ at the point (x, y) . If $T(2, 1) = 135$, and $T_x(2, 1) = 16$, and $T_y(2, 1) = -15$, estimate the temperature at the point $(2.04, 0.97)$.
26. A right circular cylinder has a radius of 50 cm and a height of 100 cm. Use differentials to estimate the change in volume of the cylinder if its height and radius are both increased by 1 cm.
27. Give the local linearization for the monthly car-loan payment function at each of the points investigated in Problem 27 on page 764.
28. In Example 3 on page 774 we found a linear approximation for $V = f(T, P)$ near $(500, 24)$. Now find a linear approximation near $(480, 20)$.
29. In Example 3 on page 774 we found a linear approximation for $V = f(T, p)$ near $(500, 24)$.
 - (a) Test the accuracy of this approximation by comparing its predicted value with the four neighboring values in the table. What do you notice? Which predicted values are accurate? Which are not? Explain your answer.

- (b) Suggest a linear approximation for $f(T, p)$ near $(500, 24)$ that does not have the property you noticed in part (a). [Hint: Estimate the partial derivatives in a different way.]
30. In a room, the temperature is given by $T = f(x, t)$ degrees Celsius, where x is the distance from a heater (in meters) and t is the elapsed time (in minutes) since the heat has been turned on. A person standing 3 meters from the heater 5 minutes after it has been turned on observes the following: (1) The temperature is increasing by 1.2°C per minute, and (2) Walking away from the heater, the temperature decreases by 2°C per meter as time is held constant. Estimate how much cooler or warmer it would be 2.5 meters from the heater after 6 minutes.
31. Van der Waal's equation relates the pressure, P , and the volume, V , of a fixed quantity of a gas at constant temperature T . For a, b, n, R constants, the equation is

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT.$$

- (a) Express P as a function of T and V .
- (b) Write a linear approximation for the change in pressure, $\Delta P = P - P_0$, resulting from a change in temperature $\Delta T = T - T_0$ and a change in pressure, $\Delta V = V - V_0$.
32. The gas equation for one mole of oxygen relates its pressure, P (in atmospheres), its temperature, T (in K), and its volume, V (in cubic decimeters, dm^3):
- $$T = 16.574 \frac{1}{V} - 0.52754 \frac{1}{V^2} - 0.3879P + 12.187VP.$$
- (a) Find the temperature T and differential dT if the volume is 25 dm^3 and the pressure is 1 atmosphere.
- (b) Use your answer to part (a) to estimate how much the volume would have to change if the pressure increased by 0.1 atmosphere and the temperature remained constant.
33. The coefficient, β , of thermal expansion of a liquid relates the change in the volume V (in m^3) of a fixed quantity of a liquid to an increase in its temperature T (in $^\circ\text{C}$):

$$dV = \beta V dT.$$

- (a) Let ρ be the density (in kg/m^3) of water as a function of temperature. (For a mass m of liquid, we have $\rho = m/V$.) Write an expression for $d\rho$ in terms of ρ and dT .

- (b) The graph in Figure 14.27 shows density of water as a function of temperature. Use it to estimate β when $T = 20^\circ\text{C}$ and when $T = 80^\circ\text{C}$.

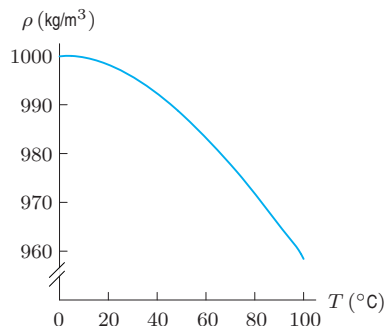


Figure 14.27

34. A fluid moves through a tube of length 1 meter and radius $r = 0.005 \pm 0.00025$ meters under a pressure $p = 10^5 \pm 1000$ pascals, at a rate $v = 0.625 \cdot 10^{-9} \text{ m}^3$ per unit time. Use differentials to estimate the maximum error in the viscosity η given by

$$\eta = \frac{\pi p r^4}{8 v}.$$

35. The period, T , of oscillation in seconds of a pendulum clock is given by $T = 2\pi\sqrt{l/g}$, where g is the acceleration due to gravity. The length of the pendulum, l , depends on the temperature, t , according to the formula $l = l_0(1 + \alpha(t - t_0))$ where l_0 is the length of the pendulum at temperature t_0 and α is a constant which characterizes the clock. The clock is set to the correct period at the temperature t_0 . How many seconds a day does the clock gain or lose when the temperature is $t_0 + \Delta t$? Show that this gain or loss is independent of l_0 .
36. Two functions that have the same local linearization at a point have contours that are tangent at this point.
- (a) If $f_x(a, b)$ or $f_y(a, b)$ is nonzero, use the local linearization to show that an equation of the line tangent at (a, b) to the contour of f through (a, b) is $f_x(a, b)(x - a) + f_y(a, b)(y - b) = 0$.
- (b) Find the slope of the tangent line if $f_y(a, b) \neq 0$.
- (c) Find an equation for the line tangent to the contour of $f(x, y) = x^2 + xy$ at $(3, 4)$.

Strengthen Your Understanding

In Problems 37–39, explain what is wrong with the statement.

37. An equation for the tangent plane to the surface $z = f(x, y)$ at the point $(3, 4)$ is

$$z = f(3, 4) + f_x(3, 4)x + f_y(3, 4)y.$$

38. If $f_x(0, 0) = g_x(0, 0)$ and $f_y(0, 0) = g_y(0, 0)$, then the surfaces $z = f(x, y)$ and $z = g(x, y)$ have the same tangent planes at the point $(0, 0)$.

39. The tangent plane to the surface $z = x^2y$ at the point $(1, 2)$ has equation

$$z = 2 + 2xy(x - 1) + x^2(y - 2).$$

In Problems 40–41, give an example of:

40. Two different functions with the same differential.

41. A surface in three space whose tangent plane at $(0, 0, 3)$ is the plane $z = 3$.

Are the statements in Problems 42–49 true or false? Give reasons for your answer.

42. The tangent plane approximation of $f(x, y) = ye^{x^2}$ at the point $(0, 1)$ is $f(x, y) \approx y$.

43. If f is a function with differential $df = 2y dx + \sin(xy) dy$, then f changes by about -0.4 between the points $(1, 2)$ and $(0.9, 2.0002)$.

44. The local linearization of $f(x, y) = x^2 + y^2$ at $(1, 1)$ gives an overestimate of the value of $f(x, y)$ at the point $(1.04, 0.95)$.

45. If two functions f and g have the same differential at the point $(1, 1)$, then $f = g$.

46. If two functions f and g have the same tangent plane at a point $(1, 1)$, then $f = g$.

47. If $f(x, y)$ is a constant function, then $df = 0$.

48. If $f(x, y)$ is a linear function, then df is a linear function of dx and dy .

49. If you zoom close enough near a point (a, b) on the contour diagram of any differentiable function, the contours will be *precisely* parallel and *exactly* equally spaced.

14.4 GRADIENTS AND DIRECTIONAL DERIVATIVES IN THE PLANE

The Rate of Change in an Arbitrary Direction: The Directional Derivative

The partial derivatives of a function f tell us the rate of change of f in the directions parallel to the coordinate axes. In this section we see how to compute the rate of change of f in an arbitrary direction.

Example 1 Figure 14.28 shows the temperature, in $^{\circ}\text{C}$, at the point (x, y) . Estimate the average rate of change of temperature as we walk from point A to point B .

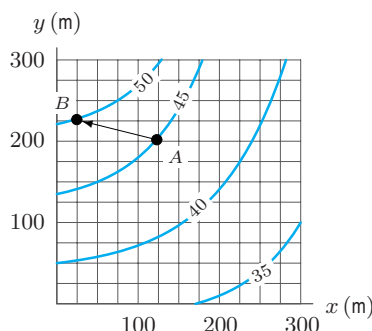


Figure 14.28: Estimating rate of change on a temperature map

Solution At the point A we are on the $H = 45^{\circ}\text{C}$ contour. At B we are on the $H = 50^{\circ}\text{C}$ contour. The displacement vector from A to B has x component approximately $-100\vec{i}$ and y component approximately $25\vec{j}$, so its length is $\sqrt{(-100)^2 + 25^2} \approx 103$. Thus, the temperature rises by 5°C as we move 103 meters, so the average rate of change of the temperature in that direction is about $5/103 \approx 0.05^{\circ}\text{C/m}$.

Suppose we want to compute the rate of change of a function $f(x, y)$ at the point $P = (a, b)$ in the direction of the unit vector $\vec{u} = u_1\vec{i} + u_2\vec{j}$. For $h > 0$, consider the point $Q = (a + hu_1, b + hu_2)$ whose displacement from P is $h\vec{u}$. (See Figure 14.29.) Since $\|\vec{u}\| = 1$, the distance from P to Q is h . Thus,

$$\begin{aligned} \text{Average rate of change} &= \frac{\text{Change in } f}{\text{Distance from } P \text{ to } Q} = \frac{f(a + hu_1, b + hu_2) - f(a, b)}{h} \\ \text{in } f \text{ from } P \text{ to } Q \end{aligned}$$